

★ WEEK 14 · LECTURE

Cause, Explanation & Fairness

Correlation vs causation, SHAP — and responsible AI

Expanded edition · 38 slides · with three hands-on labs

A model that predicts well can still be wrong, opaque, or unfair. This week we ask the harder questions: what actually CAUSES the outcome, WHY did the model decide, and is it FAIR?

Where We Are Going Today

Week 7 warned that importance is not cause. This week we take that seriously

1

Does it CAUSE, or just predict?

Correlation is not causation. Acting on a mere pattern can backfire badly.

2

WHY did the model decide that? (SHAP)

Open the black box: fairly credit each clue for a single prediction.

3

Is it FAIR?

Biased data makes biased models. How to spot and reduce unfairness.

By the end you will dissolve a fake signal, compute a Shapley value, and measure a fairness gap.

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PART 1 · CAUSE VS PREDICTION

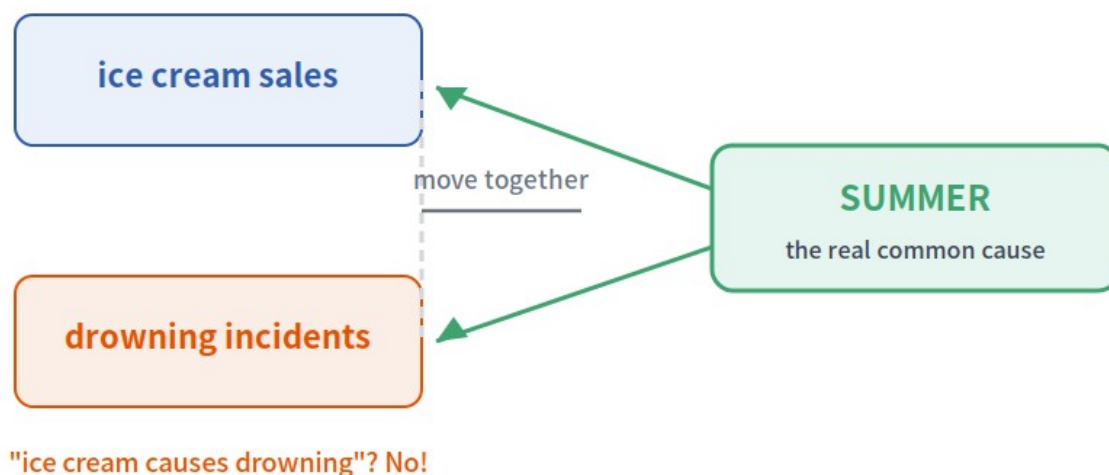
Cause vs. Prediction

Correlation is not causation

The Classic Trap

Two things move together — but one doesn't cause the other. A hidden common cause (summer) drives both.

Correlation Is Not Causation — the Hidden Cause



Why this matters in finance

A model can find a feature that PREDICTS returns without CAUSING them.

Act on a mere correlation and it can vanish the moment you trade on it.

Causal thinking asks: would changing X actually change Y?

Why Predicting Isn't Enough

Every model so far found patterns — not necessarily causes

correlation $\neq$ causation

The danger in finance

- A feature may predict returns purely by coincidence.
- Or because of a hidden third factor (like 'summer').
- Trade on it and the pattern can vanish — the factor zoo (Week 4) in a new mask.

The causal question

- Not 'does X go with Y?' but 'if I CHANGE X, will Y change?'
- Only causes survive when you act on them.
- Answering this needs special care, not just more data.

Dissolving a Fake Signal by Hand

'Social buzz predicts returns!' — ten days, one hidden regime (full version: problem set)

Comparison	Buzz-high days	Buzz-quiet days	Gap
Naive (all 10 days)	avg +1.6%	avg +0.4%	+1.2pp — 'buzz works!'
Within RALLY days only	+2.0%	+2.0%	0 — nothing
Within CALM days only	0%	0%	0 — nothing

Rallies cause both the buzz and the returns. Control for the regime and the 'signal' evaporates — 1.2pp of pure confounder.

The 'What If' Question

A causal effect compares the real world to an imagined one. The gold standard for the imagined world is a randomized A/B test.

Causal Thinking — the "What If" Question

World A: we DID act



We cut the price of a product.

Sales went up by 20%.

But would they have risen anyway?

World B: the "what if we hadn't"



The imaginary world where we DIDN'T cut.

We can never observe it directly.

Causal methods estimate it carefully.

Causal effect = World A minus World B. The gold standard for getting World B is a randomized experiment (A/B test).

How to Actually Find a Cause

One reliable way, and one careful way

$$\text{causal effect} = \text{outcome if treated} - \text{outcome if not}$$

The gold standard: experiment

- Randomly split into a treated group and a control group.
- Randomness makes the two groups otherwise identical.
- Any difference in outcome is the causal effect — the A/B test.

When you can't experiment

- In markets you often can't randomize.
- Careful causal methods mimic an experiment by controlling for confounders.
- Powerful, but easy to get wrong — handle with humility.

Three Rungs of the Causal Ladder

Each rung answers a harder question — and needs stronger evidence

Rung	The question	Evidence needed	Finance example
1 · Seeing	what goes WITH what?	any dataset (Weeks 3–8)	buzz and returns co-move
2 · Doing	what happens if I ACT?	an experiment, or careful controls	send the offer: do sales rise?
3 · Imagining	what WOULD have happened?	a causal model of the world	this trade, without my impact?

Every model so far lived on rung 1. Acting on rung-1 knowledge as if it were rung 2 is the classic quant injury.

Where Finance Can — and Cannot — Randomize

The A/B test is not off-limits; it just lives in specific rooms

You CAN randomize

- Marketing offers, credit-limit tests, product nudges.
- Execution: route half the orders via algo A, half via B.
- Chatbot replies, UI flows, research-tool rollouts.

You usually CANNOT

- Macro events, rate decisions, market-wide returns.
- Anything where the treatment leaks between groups.
- There: control for confounders, and hold conclusions loosely.

Habit: before claiming cause, ask 'could I have A/B-tested this?' — and if yes, why didn't we?

2

PART 2 · OPENING THE BOX

Opening the Black Box

SHAP: why did the model decide that?

Why a Good Score Isn't Enough

'It works' is not an acceptable answer in finance

Imagine a model rejects someone's loan. By law and by ethics, the bank must explain WHY. A black box that says 'denied, trust me' is useless — and often illegal. We need to see inside.

Why explainability matters

- Regulators require a reason for decisions.
- It builds trust with customers and risk teams.
- It helps you catch bugs and leakage (Week 10).

What we want

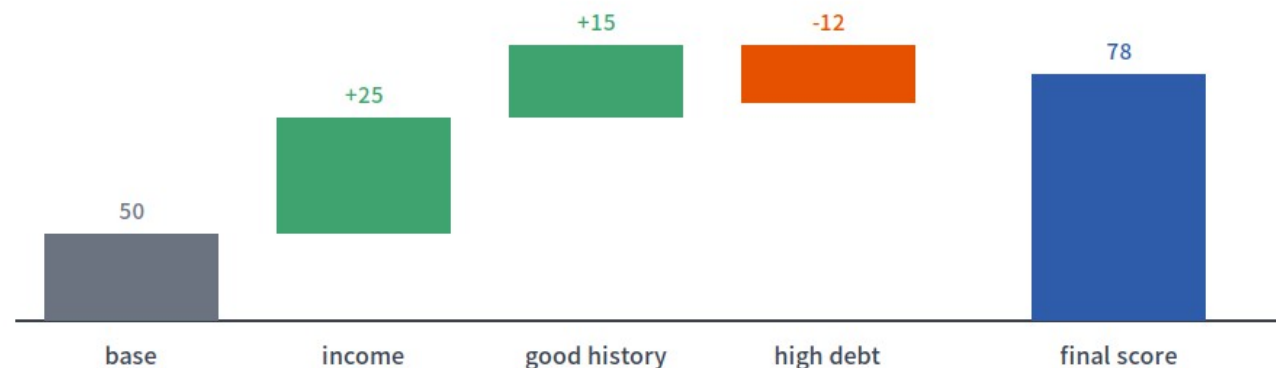
- For ONE prediction: which clues pushed it up or down?
- By how much each?
- A clear, fair breakdown. That's exactly what SHAP gives.

SHAP — Credit Each Clue Fairly

Break a single prediction into each feature's contribution — built on a fair-division idea from game theory (Shapley values).

SHAP — Fairly Credit Each Clue for the Prediction

A loan-approval score, clue by clue:



Opening the black box

Start from a base (average) score.

Each clue pushes the score up (green) or down (orange).

The pushes add up to the final prediction — fully explained.

Now you can tell a customer exactly WHY they were scored.

SHAP, as a Simple Sum

Base value, plus each clue's push, equals the prediction — nothing left unexplained

$$\text{prediction} = \text{base value} + \sum_i \phi_i$$

Reading it

- Base value = the average prediction for everyone.
- Each ϕ_i = how much one clue moved THIS prediction.
- Green pushes up, orange pushes down (from the picture).

Why it's trusted

- Mathematically fair (game theory); works on ANY model.
- Gives per-prediction AND overall importance.
- The industry standard for explaining models.

A Real Shapley Value, by Hand

Two clues, one applicant: base 0.40, $f(I) = 0.30$, $f(D) = 0.70$, $f(D, I) = 0.65$ (full version: problem set)

Clue	Arrives first: marginal push	Arrives second: marginal push	ϕ = average
Debt	$f(D) - f(\emptyset) = 0.70 - 0.40 = +0.30$	$f(D, I) - f(I) = 0.65 - 0.30 = +0.35$	+0.325
Income	$f(I) - f(\emptyset) = 0.30 - 0.40 = -0.10$	$f(D, I) - f(D) = 0.65 - 0.70 = -0.05$	-0.075
Check	base + ϕ_D + ϕ_I	0.40 + 0.325 - 0.075	= 0.65 ✓

Average each clue's push over both arrival orders — fairness by construction, and the parts sum EXACTLY to the prediction.

SHAP vs. Week 7's Feature Importance

Both rank clues — but SHAP does more

Week 7 importance (MDI/MDA)

- One number per feature for the WHOLE model.
- Tells you what matters on average.
- Can't explain a single customer's decision; can be biased (recall the warnings).

SHAP adds

- Explains ONE prediction at a time, with direction and size.
- Fairly distributed, model-agnostic.
- Still not proof of cause — explanation, not causation.

Key caution: SHAP tells you what the MODEL used, not what truly CAUSES the outcome.

Three Questions, Three Different Tools

Confusing them is the root error of this whole week

The question	The right tool	What it can NEVER tell you
What will happen?	a predictive model (W3–W8)	why, or what to change
Why did the model say that?	SHAP (today)	whether the world works that way
What actually causes it?	experiment / causal methods	— this is the top of the ladder

A bank needs all three: a score, a reason for the customer, and a lever that truly works.

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PART 3 · FAIRNESS

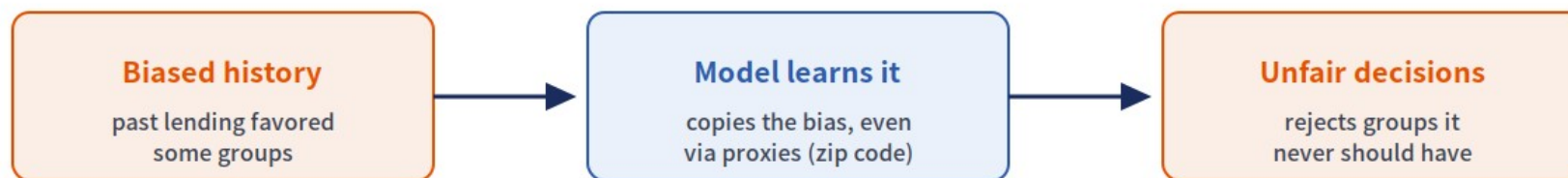
Is It Fair?

Bias, proxies, and responsible AI

Biased Data Makes Biased Models

A model learns the unfairness baked into its training history. Even after removing a protected attribute, proxies sneak it back in.

Fairness — Biased Data Makes Biased Models



The sneaky part: proxies

Removing a protected attribute isn't enough.
Zip code can secretly stand in for it,
so the bias sneaks back through the back door.

What responsible teams do

Measure outcomes ACROSS groups, not just overall accuracy.
Hunt for proxy features; use SHAP to see what drove decisions.
Keep humans in the loop for high-stakes calls.

Why Models Inherit Our Biases

The model is a mirror of its data

If past lending unfairly favored one group, the 'patterns' in that data ARE the unfairness. A model trained to copy the past will faithfully copy the discrimination, too — and at scale.

The proxy problem

- Just deleting 'race' or 'gender' isn't enough.
- Zip code, name, or shopping habits can stand in for them.
- Bias sneaks back through correlated features.

How to check

- Measure outcomes ACROSS groups, not just overall.
- Use SHAP to see if a proxy is driving decisions.
- Keep a human in the loop for high-stakes calls.

Measuring a Fairness Gap by Hand

Two groups, 100 applicants each, 80 qualified in BOTH — same talent, different treatment (full version: problem set)

	Group A	Group B	Gap
Approval rate (all)	$60 / 100 = 60\%$	$40 / 100 = 40\%$	20pp — parity gap
TPR (approved qualified)	$60 / 80 = 75\%$	$40 / 80 = 50\%$	25pp — opportunity gap
Accuracy	80%	60%	even 'accuracy' splits

Equally qualified pools, wildly different odds. One overall accuracy number would have hidden all of it — measure BY GROUP.

The Proxy Hunt

Innocent-looking features that can smuggle a protected attribute back in

Innocent-looking feature	What it can stand in for	How to catch it
Zip / postal code	neighborhood demographics	check its correlation with group labels
First name, language setting	ethnicity, national origin	SHAP: is it driving decisions?
Shopping or app-usage times	occupation, family status	group-wise outcome audits
Device model, browser	income bracket	ablate it: does the gap shrink?
School or employer lists	many protected attributes at once	domain review + human sign-off

The test is functional, not cosmetic: not 'is the column named race?' but 'does the model behave as if it were?'

Today's Story in Four Numbers

One fake signal, one fair split, two gaps

1.2pp

The buzz 'signal' —
100% confounder, 0%
cause

0.65

$= 0.40 + 0.325 - 0.075$:
a prediction, fully
accounted for

25pp

Opportunity gap
between equally-
qualified groups

3 rungs

Seeing, doing,
imagining — know
which one you're on

One discipline underneath: never let a number travel without its provenance.

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PART 4 · HANDS ON

Your Turn — Hands On

Explain a model, then check it for bias

Cause, Explain, Fair — in Our Three Apps

The hard questions show up everywhere

Lending & credit

explain + audit

SHAP tells each applicant WHY;
fairness checks compare
approval rates across groups;
hunt zip-code-style proxies.

regulated territory

Trading signals

cause or coincidence

Does the signal CAUSE returns,
or just correlate? A pattern
that's only correlation can
vanish when traded.

the factor zoo's last lesson

Fraud & chatbot

transparent alarms

SHAP explains each fraud alert
to a reviewer; fairness ensures
no group is flagged unfairly.

trust is the product

Across every app: predicting is the easy part — explaining and justifying is the job.

The Responsible-AI Habits

A short checklist for any model that affects people

1

Ask if it's a cause or a coincidence

Before acting on a pattern, ask whether changing the input would really change the outcome.

2

Explain every important decision (SHAP)

Show which clues drove each prediction. No black-box decisions on money or people.

3

Check fairness and keep humans in the loop

Measure outcomes across groups, hunt for proxies, and let people review high-stakes calls.

A model that predicts well but can't be explained or trusted is a liability, not an asset.

Three Red Lights Before Deployment

Any one of these on = the model waits, however good the score

1 · You can't explain it

SHAP gate

If nobody can say why a customer was denied, the model fails the regulator, the customer, and your own debugging.

'trust me' is not documentation

2 · Groups get different deals

fairness gate

A 25pp opportunity gap between equally-qualified groups is a lawsuit and a wrong, in either order.

audit BY GROUP, before launch

3 · The lever is unproven

causal gate

If the plan is to ACT on the pattern, rung-1 evidence isn't enough — the buzz signal was 100% confounder.

could you A/B it first?

The score answers 'can we?' — these three gates answer 'should we?'

From 'It Predicts' to 'It Deserves Trust'

The course's final ascent, week by week

Week	The claim it taught you to doubt	Today's upgrade
W1	'the backtest looks great'	...and can it be explained and defended?
W4	'this factor is significant'	significant $\neq$ causal — the ladder's rung 1
W7	'feature importance says X matters'	importance $\neq$ cause; SHAP adds per-decision 'why'
W10	'the score is honest'	honest AND explainable AND fair — three gates
W14	—	cause, explanation, fairness: the license to deploy

The arc of the course bends from 'can the model do it?' to 'can we stand behind it?'

Which Question Is This? — A 60-Second Drill

Cover the right column, answer out loud, then check yourself

Scenario	Diagnosis / tool
Signal works in-sample, dies when traded	Rung-1 correlation; likely a confounder — stratify or A/B
Customer asks why the loan was denied	SHAP — per-decision, signed, sums to the prediction
Overall accuracy 85%, but complaints from one region	Audit BY GROUP — parity and opportunity gaps
'We removed gender, so it's fair now'	Proxy problem — test behavior, not column names
'SHAP says debt causes default'	Category error — SHAP explains the MODEL, not the world

If you got 4 of 5, the responsible-AI instinct of Week 14 is yours.

Getting This Week's Data

Companion notebook — reuse the § 5 classification set

From the companion notebook (§ 5)

- Run Setup (§ 0), then the § 5 offline cell.
- Gives X (clues) and yb (label).
- We'll train a model, then EXPLAIN it. Always runs, no install.

What we'll do

- Lab 1: explain a model's clues (a SHAP-style breakdown).
- Lab 2: check accuracy across two groups (fairness).
- Real teams use the 'shap' library; we show the idea.

[CODE] FML_Data_Setup_Guide.ipynb · § 5 offline classification set

```
from sklearn.datasets import make_classification
X, yb = make_classification(n_samples=20000, n_features=20, n_informative=6,
                           weights=[0.99, 0.01], random_state=42)
# X = clues, yb = label (about 1% positive, fraud-like)
```

Hands-On Lab 1 — Explain the Clues

Which features drove the model? (a SHAP-style view)

[CODE] colab_lab1.py · data from setup guide § 5

```
import numpy as np
from sklearn.ensemble import RandomForestClassifier
from sklearn.inspection import permutation_importance
from sklearn.model_selection import train_test_split

Xtr, Xte, ytr, yte = train_test_split(X, yb, random_state=0)
m = RandomForestClassifier(n_estimators=200).fit(Xtr, ytr)

imp = permutation_importance(m, Xte, yte, n_repeats=5)
top = np.argsort(imp.importances_mean)[-5:][::-1]
for i in top:
    print(f'feature {i}:  impact {imp.importances_mean[i]:.4f}')
```

Your task (step by step)

- Train a model on X, yb (setup guide § 5).
- Use permutation importance (the shuffle test).
- Rank the clues by impact.
- Read the top few.
- Q: do the important clues make sense?

Hands-On Lab 2 — Check Fairness Across Groups

Same model, two groups — equal accuracy?

[CODE] colab_lab2.py · data from setup guide § 5

```
import numpy as np
from sklearn.metrics import accuracy_score

# pretend feature 0 marks two groups (e.g. region)
grp = (Xte[:, 0] > np.median(Xte[:, 0])).astype(int)
pred = m.predict(Xte)

for g in [0, 1]:
    mask = grp == g
    acc = accuracy_score(yte[mask], pred[mask])
    print(f'group {g}: accuracy {acc:.3f} (n={mask.sum()})')
print('a large gap between groups = fairness concern')
```

Your task (step by step)

- Split the test set into Group A and Group B.
- Measure the model's accuracy on each.
- Compare the two numbers.
- A big gap is a fairness red flag.
- Q: does the model treat both groups equally?

Hands-On Lab 3 — Claude as Your Coding Copilot

AI is allowed in this course — it explains and extends; you understand and verify

[INPUT] Prompt · Claude (paste after your Lab 1 + 2 code)

You are my machine-learning tutor.

- 1) Explain my Lab 1 and Lab 2 code above, one plain-English sentence per line.
- 2) Verify my Shapley hand-calc: base 0.40, $f(I)=0.30$, $f(D)=0.70$, $f(D,I)=0.65 \rightarrow \phi_D=0.325$, $\phi_I=-0.075$, $\text{sum} = 0.65$;
- 3) build a confounded dataset: regime drives both 'buzz' and returns; show the naive buzz-return gap and the within-regime gaps (~ 0) in one table;
- 4) fairness audit on Lab 2's model: per group, report approval rate, TPR, FPR - not just accuracy - and flag the largest gap;
- 5) three-sentence verdict: which of the three gates (explain / fair / causal) this model passes today, in this week's language.

House rules for AI use

- Allowed: explain, debug, extend code you wrote.
- Required: run it, read it, defend every line.
- Forbidden: submitting code you cannot explain.
- Tip: ask 'why', not just 'do'.

Comprehension Check

Try each question first — the answer is revealed below

Q1 Ice cream sales and drownings rise together. What's the lesson?

- A** Ice cream causes drowning
- B** Correlation can come from a hidden common cause (summer) ✓
- C** Drowning causes ice cream
- D** They are unrelated

Answer: Both are driven by a hidden common cause (summer). Correlation is not causation.

Q2 What does SHAP tell you?

- A** The true cause of an outcome
- B** How much each clue contributed to one prediction ✓
- C** The fastest model
- D** The fairest model

Answer: SHAP fairly splits a single prediction into each clue's contribution — explanation, not proof of cause.

Comprehension Check — Round Two

Two more — these cover the new slides

Q3 The buzz signal's +1.2pp gap vanished within each regime. Meaning:

- A** Buzz works only sometimes
- B** The regime confounded it — buzz caused nothing ✓
- C** We need more data
- D** The regime is a proxy

Answer: Rallies drove both buzz and returns. Stratifying by the common cause dissolved the entire 'signal'.

Q4 Both groups are 80% qualified, but TPRs are 75% vs 50%. Overall accuracy looks fine. Verdict:

- A** Fair — accuracy is what counts
- B** A 25pp opportunity gap — audit and fix before deploying ✓
- C** Groups differ in talent
- D** Delete the group column

Answer: Equal qualification, unequal odds: the aggregate hid a per-group gap. Measure by group; deleting columns won't fix behavior.

Week 14 on One Page

Photograph this slide — it is the whole lecture

Concept	One line to remember
The trap	correlation can be a hidden common cause — stratify and it may vanish.
The ladder	seeing → doing → imagining; know which rung your evidence lives on.
A/B test	randomization makes groups identical; the difference IS the cause.
SHAP	$\text{base} + \sum \phi = \text{prediction}$, exactly; fair credit per clue, per decision.
The caution	SHAP explains the MODEL; only experiments explain the world.
Fairness	measure parity and TPR BY GROUP; hunt proxies; humans own the stakes.

Next stop: the frontier — Temporal Fusion Transformers, graph neural nets, and AI agents.

Three Ideas, One Picture Each

The theme: a model that predicts well can still be wrong, opaque, or unfair. Dig deeper

1

Correlation is not causation

A pattern that predicts may not cause. Ask 'would changing X change Y?' — only causes survive when you act.

2

SHAP opens the black box

It fairly credits each clue for a single prediction (base + contributions = prediction). Explanation, not cause.

3

Fairness needs active checking

Biased data breeds biased models, even through proxies. Measure across groups; keep humans in the loop.

Keep the pictures: a summer confounder, a fairly-split prediction — and two groups on one scale.

Week 14 — Three Things to Remember

Before next class: finish Labs 1–3 and note any accuracy gap between the two groups

1

Predicting is not explaining or causing

Correlation can mislead. For a real cause, run an experiment (A/B test) or use careful causal methods.

2

SHAP explains each decision

Base value plus each clue's push equals the prediction. Fair, model-agnostic — the standard for transparency.

3

Build it responsibly

Check fairness across groups, watch for proxies, and keep humans accountable for high-stakes decisions.

Next week — The Frontier: Temporal Fusion Transformers, Graph Neural Nets, and AI Agents.